

Online appendices: Genes, Cognitive Skills, and Preferences

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Appendix A. Additional data details

TABLE A1. Factor Loadings for Trust and Reciprocities

Preference Measure	Survey Question/Statement	Factor Loading
<i>Trust</i>	When dealing with strangers, it is better to be cautious before trusting them	0.6108
	On the whole, one can trust people	-0.7818
	Nowadays one can't depend on anyone	0.7996
<i>Positive Reciprocity</i>	If someone does me a favor, I am prepared to return it	0.7161
	I go out of my way to help somebody who has been kind to me in the past	0.8319
	I am ready to assume personal costs to help somebody who helped me in the past	0.7547
<i>Negative Reciprocity</i>	If I suffer a serious wrong, I will take revenge as soon as possible, no matter	0.8720
	If somebody puts me in a difficult position, I will do the same to him/her	0.8827
	If somebody offends me, I will offend him/her back	0.7877

Notes: Responses for trust items range from 1 (Agree completely) to 4 (Disagree completely). Responses for reciprocity items range from 1 (Does not apply to me at all) to 7 (Applies to me perfectly). Higher factor scores indicate higher levels of trust or reciprocity.

TABLE A2. Definition of Variables and Response Categories

Variable	Definition	Response Categories
Risk aversion	Willingness to take risks (reverted)	0: "Not at all" to 10: "Fully prepared to take risks" (original scale reverted in our application)
Patience	General level of patience	0: "Very impatient" to 10: "Very patient"
Trust	General trust in others (Factor score based on PCA): – "On the whole, one can trust people." – "Nowadays one can't depend on anyone." – "When dealing with strangers, it is better to be cautious before trusting them."	1: "Agree completely" to 4: "Disagree completely"
Pos. Reciprocity	Willingness to return positive actions (Factor score based on PCA):	1: "Does not apply at all" to 7: "Apply completely"

Notes: Table A2 displays outcomes, together with the definition and the response categories.

TABLE A3. Definition of control variables and response Categories

Variable	Definition	Response Categories
PGI CS	Polygenic index for cognitive skills	Numerical scale, standardized to have mean zero and standard deviation one
Female	Indicator for being female	0: "Male"; 1: "Female"
Year of birth	Year of respondent's birth	Numeric variable
Age	Age, defined as the difference between survey year and year of birth from the year when outcome was observed	Numerical variable
Survey year	Year when interview was conducted	Numerical variable
Principal components of genetic data	First ten principal components of the respondent's genetic data	Numerical scale, standardized to have mean zero and standard deviation one
Paternal education	Father's level of education	1: "Do not know"; 2: "Secondary School"; 3: "Intermediate School"; 4: "Technical School"; 5: "Upper Secondary School"; 6: "Other Degree"; 7: "No School Degree"
Maternal education	Mother's level of education	1: "Do not know"; 2: "Secondary School"; 3: "Intermediate School"; 4: "Technical School"; 5: "Upper Secondary School"; 6: "Other Degree"; 7: "No School Degree"
Paternal occupational status	Father's occupational status when respondent was 15 years old	1: "Missing"; 2: "Not Employed"; 3: "Self-Employed"; 4: "Blue-Collar"; 5: "White-Collar"; 6: "Civil Service"
Maternal occupational status	Mother's occupational status when respondent was 15 years old	1: "Missing"; 2: "Not Employed"; 3: "Self-Employed"; 4: "Blue-Collar"; 5: "White-Collar"; 6: "Civil Service"
Parents separated	Indicator that is equal to 3 if respondent lived her first 15 years in the same household as biological parents and 2 otherwise	1: "Missing"; 2: "Other types of Household"; 3: "Two-Parents Household"

Notes: Table A3 displays control variables, together with their definitions and response categories.

TABLE A4. Covariates balance

	PGI CS < Median		PGI CS ≥ Median		Std. Diff.
	Mean or N	SD or (%)	Mean or N	SD or (%)	
<i>Gender</i>					
Male	515	44.8	535	46.6	.0349
Female	634	55.2	614	53.4	
<i>Birth year</i>	1,965	19	1,963	19.3	.12
<i>Father's education</i>					
Upper secondary sch.	127	11.1	208	18.1	.223
Intermediate sch.	186	16.2	167	14.5	
Technical sch.	9	.783	8	.696	
Secondary sch.	636	55.4	617	53.7	
No school degree	26	2.26	29	2.52	
Other degree	26	2.26	20	1.74	
Missing	139	12.1	100	8.7	
<i>Mother's education</i>					
Upper secondary sch.	71	6.18	124	10.8	.198
Intermediate sch.	232	20.2	236	20.5	
Technical sch.	8	.696	14	1.22	
Secondary sch.	680	59.2	645	56.1	
No school degree	41	3.57	34	2.96	
Other degree	19	1.65	24	2.09	
Missing	98	8.53	72	6.27	
<i>Father's occupation</i>					
Self-employed	114	9.92	155	13.5	.229
White-collar	333	29	344	29.9	
Civil service	66	5.74	112	9.75	
Blue-collar	434	37.8	365	31.8	
Not employed	103	8.96	104	9.05	
Missing	99	8.62	69	6.01	
<i>Mother's occupation</i>					
Self-employed	61	5.31	74	6.44	.142
White-collar	354	30.8	400	34.8	
Civil service	16	1.39	25	2.18	
Blue-collar	235	20.5	190	16.5	
Not employed	372	32.4	358	31.2	
Missing	111	9.66	102	8.88	
<i>HH. env. growing up</i>					
Two-parents HH.	836	72.8	838	72.9	.0668
Other types of HH.	264	23	247	21.5	
Missing	49	4.26	64	5.57	

Notes: Table A4 displays the standardized differences in covariates between individuals with polygenic index for cognitive skills above and below the median. A standardized difference greater than 0.1 in absolute value could indicate covariate imbalance between the two groups, as proposed by Austin (2001).

Source: Authors' calculations based on SOEP-IS 2020.

Appendix B. Details on the sensitivity analyses

B.1. Relative importance of selection on unobservables consistent with a null effect

The key insight of Oster’s delta method is that the degree of selection on observables can inform the degree of selection on unobservables. The proportionality factor δ is defined as:

$$(A1) \quad \delta = \frac{\sigma_{PGI_{CP}U}}{\sigma_{PGI_{CP}O}},$$

where $\sigma_{PGI_{CP}U}$ represents the covariance between the treatment and unobserved confounders, u , and $\sigma_{PGI_{CP}O}$ represents the covariance between the treatment and observed controls, o , which in our case contains our controls for age, year of birth, parental controls and genetic PCs.

To assess robustness, (Oster 2019) derives the bias-adjusted treatment effect:

$$(A2) \quad \beta_1^* = \beta_{1,\text{controlled}} - \delta \left(\beta_{1,\text{uncontrolled}} - \beta_{1,\text{controlled}} \right) \times \frac{R_{\text{max}} - R_{\text{controlled}}}{R_{\text{controlled}} - R_{\text{uncontrolled}}}.$$

Here, R_{max} represents the hypothetical R^2 if all confounders were observed, $R_{\text{controlled}}$ is the R^2 with observed controls, and $R_{\text{uncontrolled}}$ is the R^2 without controls. Typically, R_{max} is set to $1.3 \cdot R_{\text{controlled}}$ (Oster 2019). The intuition is that typically, data in social science is gathered to explain the phenomena we are interested in. In order to understand the selection based on unobservables necessary to be consistent with the absence of no effect, *i.e.*, $\beta_1^* = 0$, we set Equation A2 equal to zero and solve for δ . The results are displayed in Table A5 and A6.

TABLE A5. Relative importance of selection on unobservables - risk and time preferences

	Risk aversion			Smoking			Patience			Fin. assets		
	(1) Base	(2) Full		(3) Base	(4) Full		(5) Base	(6) Full		(7) Base	(8) Full	
PGICS	0.058*** (0.020)	0.050** (0.021)		-0.048*** (0.009)	-0.035*** (0.009)		0.008 (0.021)	0.003 (0.023)		0.049*** (0.010)	0.034*** (0.011)	
Delta		10.662			5.399			1.987			4.615	
Observations	2298	2298		2057	2057		2200	2200		2256	2256	
R ²	0.003	0.138		0.013	0.143		0.000	0.061		0.010	0.113	

Notes: Table A5 displays the relative importance of selection on unobservables, relative to observables, needed to fully explain the estimates in our main specification, detailed in Equation 4. The outcomes correspond to our set of risk and time preference measures. The odd columns display the effect estimate of a regression of the outcome on the PGI CS. The even columns display the results of our main specification. Delta describes the relative importance of selection on unobservables, relative to selection on observables, to fully explain our effect. Consistent with the recommendation by Oster (2019), we set the R^2 associated with a model, that includes all unobservables, equal to 1.3 times the R^2 in the corresponding even column. Oster (2019) suggests that deltas larger than one can be considered robust. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

TABLE A6. Relative importance of selection on unobservables - social preferences

	Trust		Pos. reciprocity		Neg. reciprocity		Altruism		Lending help		Lending belongings		Lending money	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	Base	Full	Base	Full	Base	Full	Base	Full	Base	Full	Base	Full	Base	Full
PGICS	0.131***	0.092***	-0.030	-0.025	-	-	0.011	0.016	-0.022	-0.009	0.049**	0.053**	0.037*	0.055***
	0.066*** 0.058***													
	(0.021)	(0.022)	(0.023)	(0.024)	(0.021)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.021)	(0.022)	(0.021)	(0.021)
Delta	4.432	2189	2197	2197	10.330	10.823	-13.770	1.889	2173	2166	2166	152.571	2167	-12.498
Observations	2189	2189	2197	2197	2197	2197	2130	2130	2173	2173	2166	2166	2167	2167
R ²	0.017	0.118	0.001	0.053	0.004	0.079	0.000	0.104	0.000	0.130	0.002	0.140	0.001	0.211

Notes: Table A6 displays the relative importance of selection on unobservables, relative to observables, needed to fully explain the estimates in our main specification, detailed in Equation 4. The outcomes correspond to our set of social preference measures. The odd columns display the effect estimate of a regression of the outcome on the PGI CS. The even columns display the results of our main specification. Delta describes the relative importance of selection on unobservables, relative to selection on observables, to fully explain our effect. Consistent with the recommendation by Oster (2019), we set the R^2 associated with a model, that includes all unobservables, equal to 1.3 times the R^2 in the corresponding even column. Oster (2019) suggests that deltas larger than one can be considered robust. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Source: Authors' calculations based on SOEP-IS 2020.

B.2. Accounting for individual level heterogeneity

In our baseline specification, we estimate the model using ordinary least squares (OLS) as follows:

$$(A3) \quad y_{it} = \beta_0 + \beta_1 PGI\ CS_{it} + \beta_2 X_{it} + \beta_3 Z_{it} + \beta_4 PC_{it} + \epsilon_{it}.$$

This model assumes that the error term ϵ_{it} captures all unobserved factors affecting the outcome y_{it} .

To address potential unobserved, time-invariant heterogeneity that might influence the outcome but is not captured by our observed covariates, we augment our model by introducing a random effect. Specifically, we allow for an individual-specific component, u_i , which captures persistent characteristics unique to each individual. The augmented model can be written as:

$$(A4) \quad y_{it} = \alpha_0 + \alpha_1 PGI\ CS_{it} + \alpha_2 X_{it} + \alpha_3 Z_{it} + \alpha_4 PC_{it} + u_i + \mu_{it},$$

where:

- u_i is the individual-specific random effect, which we assume to be a random draw from a normal distribution, that is,

$$u_i \sim \mathcal{N}(0, \sigma_u^2),$$

capturing all time-invariant unobserved heterogeneity;

- ϵ_{it} is the idiosyncratic error term.

By specifying the model in this way, we allow the intercept to vary across individuals. The individual-specific effect u_i captures unobserved factors such as family background, ancestry effects or even genetic endowments of the parents, among others. This random effects model thus helps to mitigate bias due to omitted time-invariant variables.

The efficiency of the random effects estimator relies on the assumption that the individual-specific effects u_i are uncorrelated with the explanatory variables. Our sensitivity analysis involves estimating the random effects model in Equation (A4) and comparing the resulting coefficient estimates – particularly for β_1 , the coefficient on the PGI for cognitive performance – with those from the OLS specification in Equation (A3). If the estimates remain stable across these specifications, it provides evidence that unobserved heterogeneity does not threaten the validity of our empirical findings.

TABLE A7. Random effects model for risk/time preferences

	(1) Risk aversion	(2) Smoking	(3) Patience
PGI CS	0.051*** (0.015)	-0.034*** (0.009)	0.000 (0.020)
Observations	17319	2904	3665
Clusters	2298	2057	2200

Notes: Table A16 displays the association between PGI score for cognitive skills and risk/time preference measures based on random effects model. The results for the probability of holding financial assets are omitted due to the lack of panel data for the outcome. The regression controls for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender and birth year. Cluster robust standard errors are included in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

TABLE A8. Random effects model for social preferences

	(1) Trust	(2) Pos. reciprocity	(3) Neg. reciprocity	(4) Altruism	(5) Lending help	(6) Lending belongings	(7) Lending money
PGI CS	0.098*** (0.019)	-0.017 (0.019)	-0.065*** (0.019)	0.007 (0.019)	-0.028 (0.017)	0.052** (0.021)	0.056*** (0.020)
Observations	3768	4295	4297	3275	6653	2423	2421
Clusters	2194	2200	2200	2130	2173	2166	2167

Notes: Table A8 displays the association between PGI score for cognitive skills and social preference outcomes based on random effects model. The regression controls for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender and birth year. Cluster robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

B.3. Addressing Pleiotropy through Sensitivity Analysis with Non-Cognitive PGIs

Genetic variants may influence multiple phenotypes, a phenomenon known as *pleiotropy* (Benjamin et al. 2024). In our context, pleiotropy implies that some SNPs contributing to the PGI CP may also affect non-cognitive traits. This overlapping genetic influence can potentially confound the estimated association between the CP PGI and the outcome of interest if part of the observed effect is actually driven by genetic influences on other correlated traits.

Furthermore, to the extent that the SNPs informing the PGIs for non-cognitive skills are in linkage disequilibrium with those contributing to the CP PGI, the inclusion of these additional PGIs also helps to account for any correlation arising from shared genetic architecture. In essence, our sensitivity analysis not only adjusts for potential pleiotropic influences but also mitigates any bias introduced by linkage disequilibrium between the SNPs underlying different PGIs.

To address these concerns, we implement a sensitivity analysis by including PGIs for non-cognitive traits – specifically, openness to experience, extraversion, and neuroticism – in our main empirical model. These traits are selected because they capture key dimensions of personality that are commonly associated with both cognitive performance and social outcomes and are included in the SOEP-G (Koellinger et al. 2023). By controlling for these additional PGIs, we aim to isolate the unique effect of the CP PGI, ensuring that the observed association is not spuriously driven by pleiotropic effects.

Tables A9 and A10 display the results from models that incorporate these non-cognitive PGIs.

TABLE A9. Addressing pleiotropy in the association between PGI CS and risk/time preferences

	(1) Risk aversion	(2) Smoking	(3) Patience	(4) Fin. assets
PGI CS	0.053*** (0.021)	-0.032*** (0.010)	-0.006 (0.023)	0.034*** (0.011)
Observations	2298	2057	2200	2256
R ²	0.143	0.146	0.064	0.114

Notes: Table A9 displays the association between PGI score for cognitive skills and risk/time preference outcomes after controlling for PGIs related to non-cognitive traits to address pleiotropy concerns. In addition to the full specification, single-trait PGIs for extraversion, neuroticism, and openness, dimensions of the Big Five personality traits for which genetic data is available, are included. The full specification controls include age, age squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender, and birth year. Robust standard errors are provided in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

TABLE A10. Addressing pleiotropy in the association between PGI CS and social preferences

	(1) Trust	(2) Pos. reciprocity	(3) Neg. reciprocity	(4) Altruism	(5) Lending help	(6) Lending belongings	(7) Lending money
PGI CS	0.089*** (0.022)	-0.025 (0.025)	-0.051** (0.022)	0.018 (0.023)	-0.007 (0.022)	0.048** (0.022)	0.051** (0.020)
Observations	2189	2197	2197	2130	2173	2166	2167
R ²	0.120	0.053	0.082	0.106	0.131	0.144	0.212

Notes: Table A10 displays the association between PGI score for cognitive skills and social preference outcomes after controlling for PGIs related to non-cognitive traits to address pleiotropy concerns. In addition to the full specification, single-trait PGIs for extraversion, neuroticism, and openness, dimensions of the Big Five personality traits for which genetic data is available, are included. The full specification controls include age, age squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender, and birth year. Robust standard errors are provided in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

B.4. Multiple Hypothesis Testing Adjustments

In our empirical application, multiple hypothesis testing presents a challenge in maintaining statistical validity. When testing multiple hypotheses, the probability of false positives increases, necessitating adjustments to p-values. In this section we apply three widely used procedures: the Bonferroni adjustment (Dunn 1961), Westfall and Young (1992)'s resampling-based approach, and the adjustment proposed by List, Shaikh, and Xu (2019).

Bonferroni Adjustment. The Bonferroni adjustment is one of the most widely used methods to control the family-wise error rate (FWER). It achieves this by setting a more stringent significance threshold:

$$(A5) \quad p^{\text{adjusted}} = \min(1, m \cdot p)$$

where m is the number of hypotheses tested, and p is the unadjusted p-value. While conservative, this method ensures strong control of Type I error at the cost of increased Type II errors.

Westfall and Young (1992)'s Resampling-Based Approach. Westfall and Young (1992) introduce a resampling-based method that accounts for the dependency structure among tests. Instead of applying a fixed correction factor, they use permutation techniques to estimate the joint distribution of p-values, allowing for a less conservative adjustment while still controlling the FWER. This method is particularly useful when test statistics are correlated, reducing the loss of statistical power compared to the Bonferroni adjustment.

List, Shaikh, and Xu (2019)'s Adjustment for Experimental Economics. List, Shaikh, and Xu (2019) propose an approach, mainly tailored to experimental economics, that balances statistical power and error control. Their method builds upon family-wise and false discovery rate (FDR) adjustments, recognizing that in experimental economics, researchers often face a large number of simultaneous tests. Their approach suggests a combination of stepwise adjustments and economic theory-based pre-registration to mitigate concerns about data mining.

Comparison of Methods. These three methods offer different trade-offs:

- **The Bonferroni adjustment** is simple and ensures strong control over false positives but is overly conservative when hypotheses are correlated.
- **Westfall and Young (1992)** provide an improved alternative that maintains control over false positives while increasing statistical power through resampling techniques.
- **List, Shaikh, and Xu (2019)** offer a more flexible approach for experimental economics, allowing for more nuanced adjustments based on empirical settings and research design.

We apply these procedures to our hypotheses testing for different groups of hypotheses. The results are displayed in Table A11 and A12.

TABLE A11. Multiple hypothesis testing

Hyp. num.	Dep. var.	Coeff.	P-value	List et al.	Bonferroni	Westfall-Young	Randomization
1	Risk aversion	0.050	0.016	0.070	0.163	0.086	0.014
2	Smoking	-0.035	0.000	0.000	0.002	0.003	0.000
3	Patience	0.003	0.880	0.883	1.000	0.899	0.879
4	Fin. assets	0.034	0.002	0.034	0.044	0.016	0.003
5	Trust	0.092	0.000	0.000	0.002	0.000	0.000
6	Pos. reciprocity	-0.025	0.298	0.759	1.000	0.759	0.298
7	Neg. reciprocity	-0.058	0.008	0.061	0.101	0.058	0.010
8	Altruism	0.016	0.468	0.859	1.000	0.838	0.465
9	Lending help	-0.009	0.693	0.898	1.000	0.899	0.703
10	Lending belongings	0.053	0.017	0.082	0.158	0.086	0.016
11	Lending Money	0.055	0.008	0.057	0.084	0.058	0.006
All			0.002				

Notes: Table A11 displays the original coefficients and p-values of PGI CS and the adjusted p-values from different methods under different outcomes. The adjustment procedure by List et al. (2019) and Westfall-Young allows for dependencies across outcomes. Bonferroni method restricts an upper bound to the adjusted p-values at 1. Randomization is based on randomised inference as in Young (2019). Statistical inference is based on randomised inference as in Young (2019), where a multiple hypothesis testing across all equations is performed. *Source:* Authors' calculations based on SOEP-IS 2020.

TABLE A12. Multiple hypothesis testing by preference domains

Hyp. num.	Dep. var.	Coeff.	P-value	List et al.	Bonferroni	Westfall-Young	Randomi- zation
Family 1: Risk preferences							
1	Risk aversion	0.050	0.016	0.015	0.030	0.016	0.014
2	Smoking	-0.035	0.000	0.000	0.000	0.001	0.000
All			0.000				
Family 2: Time preferences							
1	Patience	0.003	0.880	0.882	1.000	0.881	0.879
2	Fin. assets	0.034	0.002	0.003	0.003	0.003	0.002
All			0.005				
Family 3: Social preferences							
1	Trust	0.092	0.000	0.000	0.001	0.000	0.000
2	Pos. reciprocity	-0.025	0.298	0.644	1.000	0.661	0.298
3	Neg. reciprocity	-0.058	0.008	0.043	0.063	0.043	0.010
4	Altruism	0.016	0.468	0.707	1.000	0.706	0.465
5	Lending help	-0.009	0.693	0.696	1.000	0.706	0.703
6	Lending belongings	0.053	0.017	0.066	0.119	0.065	0.016
7	Lending money	0.055	0.008	0.048	0.059	0.043	0.006
All			0.000				

Notes: Table A12 displays the original coefficients and p-values of PGI CS and the adjusted p-values from different methods under different outcomes according to preference domains. The adjustment procedure by List et al. (2019) and Westfall-Young allows for dependencies across outcomes. Bonferroni method restricts an upper bound to the adjusted p-values at 1. Statistical inference is based on randomised inference as in Young (2019), where a multiple hypothesis testing across all equations is performed. *Source:* Authors' calculations based on SOEP-IS 2020.

B.5. Quality Control Procedures: Mild-QC vs. Strict-QC

In the SOEP-G, two distinct quality control (QC) pipelines were implemented for processing the genetic data:¹

- **Mild-QC:** This pipeline applies less stringent criteria. It excludes individuals with extreme values – such as those with per-chromosome missingness greater than 50% and marked heterozygosity/homozygosity outliers – while retaining a larger sample size.
- **Strict-QC:** In contrast, the strict-QC pipeline enforces rigorous thresholds by excluding individuals with per-chromosome missingness over 20% and those with excessive heterozygosity/homozygosity. This results in a smaller, but arguably higher-quality dataset.

Despite the stricter criteria, both pipelines yield PGIs with nearly equivalent predictive accuracy (Koellinger et al. 2023). The usage of both quality standards comes with advantages and disadvantages:

- Using the **mild-QC** data when maximizing sample size (and thus statistical power) is essential, especially for analyses that rely on aggregated PGIs.
- Opting for the **strict-QC** data for studies that require very high genotype fidelity, such as analyses of single genetic variants or detailed genotype-phenotype associations.

In our main analyses, we relied on the sample that results from the mild-QC. In Table A13 and A14, we present results based on the sample that results from the strict-QC.

¹See (Koellinger et al. 2023) for details.

TABLE A13. Association between PGI CS and risk/time preferences under strict quality control

	(1) Risk aversion	(2) Smoking	(3) Patience	(4) Fin. assets
PGI CS	0.045** (0.021)	-0.027*** (0.010)	-0.001 (0.024)	0.029*** (0.011)
Observations	2092	1867	2001	2054
R ²	0.136	0.150	0.065	0.112

Notes: Table A13 displays the results for risk/time preferences upon imposing strict quality control on the PGI score for cognitive skills. The associations shown are obtained from regressions with full specification, controlling for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender and birth year. Robust standard errors are reported in parentheses. * p < 0.10, ** p < 0.05, and *** p < 0.01. *Source:* Authors' calculations based on SOEP-IS 2020.

TABLE A14. Association between PGI CS and social preferences under strict quality control

	(1) Trust	(2) Pos. reciprocity	(3) Neg. reciprocity	(4) Altruism	(5) Lending help	(6) Lending belongings	(7) Lending money
PGI CS	0.091*** (0.023)	-0.029 (0.025)	-0.055** (0.023)	0.015 (0.024)	-0.012 (0.023)	0.039* (0.023)	0.051** (0.021)
Observations	1991	1998	1998	1937	1975	1972	1973
R ²	0.117	0.064	0.077	0.106	0.132	0.145	0.222

Notes: Table A14 displays the results for social preferences upon imposing strict quality control on the PGI score for cognitive skills. The associations shown are obtained from regressions with full specification, controlling for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender and birth year. Robust standard errors are reported in parentheses. * p < 0.10, ** p < 0.05, and *** p < 0.01. *Source:* Authors' calculations based on SOEP-IS 2020.

B.6. Motivation and Procedure for Controlling Measurement Error in PGI Regressions

PGIs are constructed as weighted sums of allele counts and serve as predictors for various phenotypes. However, because the weights are estimated from finite-sample GWAS summary statistics, a PGI is an *unbiased but noisy* measure of the true *additive SNP factor* – the optimal linear predictor based on all measured SNPs. This noise induces *errors-in-variables bias* in regression analyses that use the PGI as an explanatory variable, typically attenuating estimated coefficients toward zero.

To address this issue, the authors derive a measurement-error correction (also known as regression disattenuation) estimator. The key steps are:

- (a) **Characterization of Measurement Error:** The PGI is shown to be equivalent to the true additive SNP factor plus classical measurement error. In particular, if we denote the true factor as g_i and the PGI as $\hat{g}_i$, then

$$\hat{g}_i = g_i + e_i,$$

where e_i is mean-zero error that is uncorrelated with g_i . The magnitude of the measurement error is captured by a factor

$$\rho = \sqrt{\frac{h_{\text{SNP}}^2}{R^2}},$$

with h_{SNP}^2 representing the SNP heritability (*i.e.*, the maximum predictive power of g_i) and R^2 being the actual variance explained by the PGI.

- (b) **Bias in Standard Regressions:** When the PGI is used as a proxy for the unobserved g_i in a regression model, the estimated coefficients are biased due to measurement error. In the simplest univariate case, the estimated slope is attenuated by a factor of $1/\rho$.
- (c) **Corrected Estimator:** Becker et al. (2021) extend the standard errors-in-variables framework to the multivariate context. They derive an estimator that “inverts” the attenuation bias by adjusting the regression coefficients using estimates of the variance-covariance structure of the variables and the measurement error parameter ρ . The essential idea is to replace the naive OLS estimates with corrected coefficients that reflect the true effect of the standardized additive SNP factor.

(d) **Practical Implementation:** In practice, the procedure involves:

- Estimating h_{SNP}^2 (using methods such as Genome-based restricted maximum likelihood (GREML) or BOLT-REML) and R^2 from regressions of the phenotype on the PGI.
- Computing ρ , and then using it to adjust the estimated regression coefficients and standard errors.
- In the univariate case, this amounts to multiplying the OLS coefficient and its standard error by ρ , while in the multivariate case, the adjustment is applied via a matrix correction.

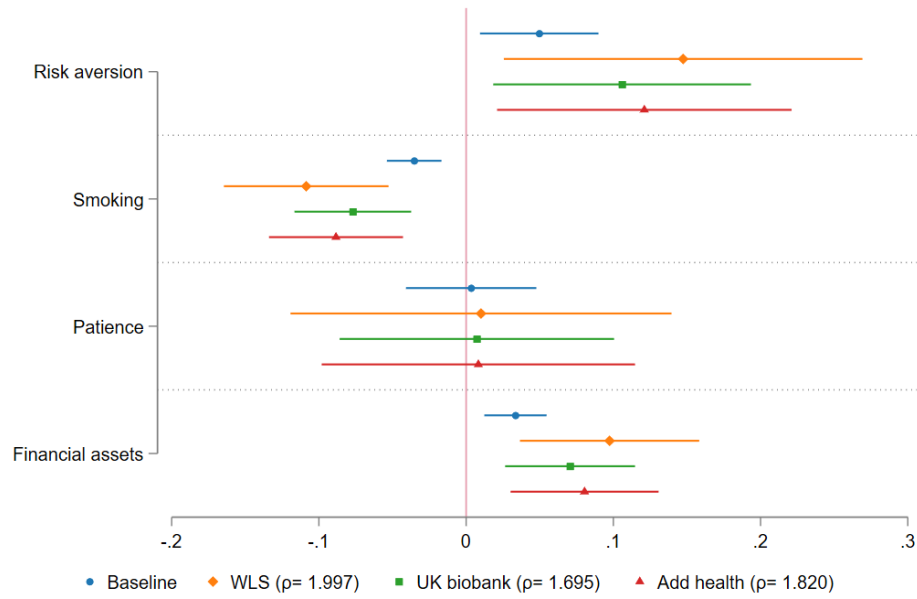
The corrected coefficients are reported in units of the true additive SNP factor, providing more interpretable and comparable effect sizes across studies and different PGIs.

This procedure enhances the validity of empirical inferences by mitigating the downward bias caused by measurement error in the PGI.

In order to understand the extent of attenuation bias in our main results, we apply the measurement error correction to our main estimates. Ideally, we would like to have access to h_{SNP}^2 and R^2 from a GWAS in SOEP-G. In absence of these output, we rely on available inputs from the Wisconsin Longitudinal Study (WLS), UK Biobank, and Add Health. Input from the WLS and UK Biobank comes from Becker et al. (2021). The inputs from Add Health are based on Sanz-de Galdeano and Terskaya (2023).² The results are displayed in Figure A1 and A2. In each of these figures, we show the estimate based on our main analysis together with the ones which are based on the measurement error corrected versions.

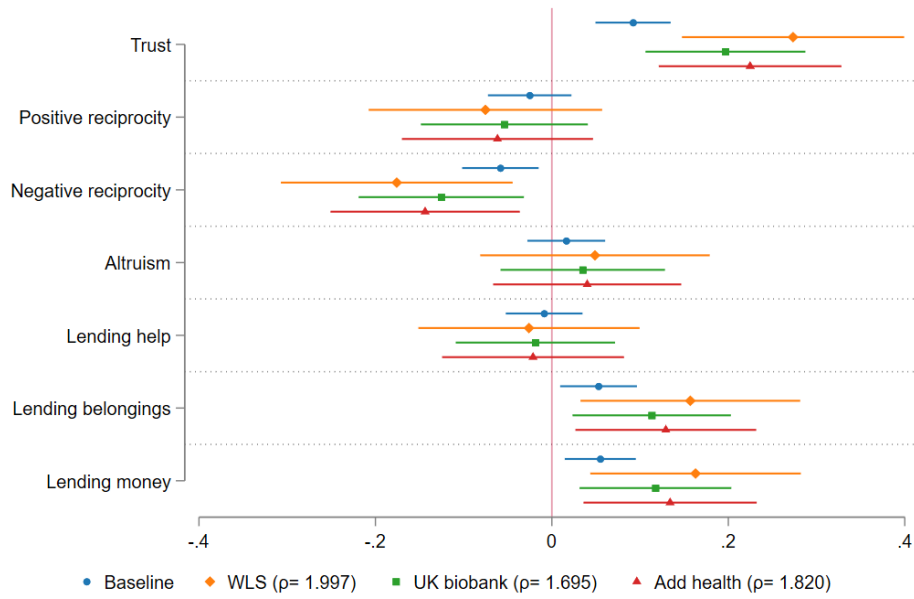
²The Python code for the measurement error correction is provided by Becker et al. (2021) via https://github.com/JonJala/pgi_correct. This version is associated with the commit hash 12d46d7.

FIGURE A1. The effect of the measurement-error-corrected PGI CP on risk aversion and patience(Becker et al. 2021)



Notes: Figure A1 illustrates the association between PGI CP and risk aversion as well as patience based on our main specification, alongside the corresponding associations using the measurement-error-corrected PGI CP. For each outcome, results are shown for h^2_{SNP} and R^2 from the WLS, UK Biobank, and Add Health (Becker et al. 2021; Sanz-de Galdeano and Terskaya 2023). In all specifications, controls include gender, age, year of birth, parental school leaving degrees, parental occupational status, an indicator for parental separation, and the first ten genetic principal components. The measurement-error-corrected analyses employ the method of Becker et al. (2021). Throughout, 95% confidence intervals are reported based on analytic, robust standard errors.

FIGURE A2. The effect of the measurement-error-corrected PGI CP on social preferences(Becker et al. 2021)



Notes: Figure A1 illustrates the association between PGI CP and social preferences based on our main specification, alongside the corresponding associations using the measurement-error-corrected PGI CP. For each outcome, results are shown for h^2_{SNP} and R^2 from the WLS, UK Biobank, and Add Health (Becker et al. 2021; Sanz-de Galdeano and Terskaya 2023). In all specifications, controls include gender, age, year of birth, parental school leaving degrees, parental occupational status, an indicator for parental separation, and the first ten genetic principal components. The measurement-error-corrected analyses employ the method of Becker et al. (2021). Throughout, 95% confidence intervals are reported based on analytic, robust standard errors.

TABLE A15. Measurement error correction

	Main results $\rho = 0$	WLS $\rho = 1.997$	Add health $\rho = 1.820$	UK biobank $\rho = 1.695$
Risk aversion	0.050** (0.021)	0.147** (0.062)	0.121** (0.051)	0.106** (0.045)
Smoking	-0.035*** (0.009)	-0.109*** (0.029)	-0.088*** (0.023)	-0.077*** (0.020)
Patience	0.003 (0.023)	0.010 (0.066)	0.008 (0.054)	0.007 (0.048)
Fin. assets	0.034*** (0.011)	0.097*** (0.031)	0.080*** (0.026)	0.071*** (0.023)
Trust	0.092*** (0.022)	0.273*** (0.064)	0.225*** (0.053)	0.197*** (0.046)
Pos. reciprocity	-0.025 (0.024)	-0.075 (0.068)	-0.062 (0.055)	-0.054 (0.048)
Neg. reciprocity	-0.058*** (0.022)	-0.176*** (0.067)	-0.144*** (0.055)	-0.125*** (0.048)
Altruism	0.016 (0.022)	0.049 (0.066)	0.040 (0.054)	0.035 (0.048)
Lending help	-0.009 (0.022)	-0.026 (0.064)	-0.021 (0.053)	-0.019 (0.046)
Lending belongings	0.053** (0.022)	0.157** (0.064)	0.129** (0.052)	0.113** (0.046)
Lending money	0.055*** (0.021)	0.163*** (0.061)	0.134*** (0.050)	0.117*** (0.044)

Notes: Table A15 displays the original coefficients PGI CS coefficients preference outcomes and the corresponding measurement-error adjusted statistics computed using the method by Becker et al. (2021). Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

B.7. Checking for non-linearities

TABLE A16. Non-Linearity between PGI CS and risk/time preferences

	(1) Risk aversion	(2) Smoking	(3) Patience	(4) Fin. assets
PGI CS	0.051** (0.021)	-0.036*** (0.010)	0.002 (0.023)	0.033*** (0.011)
PGI CS squared	-0.010 (0.014)	0.008 (0.006)	0.014 (0.015)	0.001 (0.007)
Observations	2298	2057	2200	2256
R ²	0.139	0.144	0.061	0.113

Notes: Table A16 displays the results from examining linearity in the association between PGI score for cognitive skills and risk/time preference measures based on OLS regressions. The regression controls for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender, birth year. Robust standard errors are included in parentheses. * p < 0.10, ** p < 0.05, and *** p < 0.01. Source: Authors' calculations based on SOEP-IS 2020.

TABLE A17. Non-Linearity between PGI CS and social preferences

	(1) Trust	(2) Pos. reciprocity	(3) Neg. reciprocity	(4) Altruism	(5) Lending help	(6) Lending belongings	(7) Lending money
PGI CS	0.089*** (0.022)	-0.024 (0.025)	-0.060*** (0.023)	0.018 (0.023)	-0.010 (0.022)	0.050** (0.022)	0.054*** (0.020)
PGI CS squared	0.029** (0.014)	-0.011 (0.018)	0.012 (0.015)	-0.011 (0.016)	0.013 (0.015)	0.026* (0.015)	0.005 (0.015)
Observations	2189	2197	2197	2130	2173	2166	2167
R ²	0.120	0.053	0.079	0.104	0.130	0.142	0.211

Notes: Table A17 displays the results from examining linearity in the association between PGI score for cognitive skills and social preference outcomes based on OLS regressions. The regression controls for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender, birth year. Robust standard errors are reported in parentheses. * p < 0.10, ** p < 0.05, and *** p < 0.01. Source: Authors' calculations based on SOEP-IS 2020.

B.8. Additional controls for revealed preferences outcomes

TABLE A18. Association between PGI CS and revealed preferences controlling for education and income

	(1) Smoking	(2) Fin. assets	(3) Lending help	(4) Lending belongings	(5) Lending money
Panel A: Main specification controls					
PGI CS	-0.035*** (0.009)	0.034*** (0.011)	-0.009 (0.022)	0.053** (0.022)	0.055*** (0.021)
R^2	0.143	0.113	0.130	0.140	0.211
Adjusted R^2	0.092	0.067	0.082	0.093	0.168
Panel B: Main specification and education controls					
PGI CS	-0.023** (0.010)	0.022* (0.011)	-0.004 (0.022)	0.038* (0.022)	0.046** (0.021)
R^2	0.154	0.125	0.128	0.150	0.208
Adjusted R^2	0.101	0.077	0.079	0.101	0.163
Panel C: Main specification and income controls					
PGI CS	-0.031*** (0.010)	0.023** (0.011)	-0.012 (0.023)	0.057** (0.022)	0.060*** (0.021)
R^2	0.169	0.154	0.135	0.147	0.207
Adjusted R^2	0.118	0.108	0.086	0.098	0.161
Panel D: Main specification and both additional controls					
PGI CS	-0.022** (0.010)	0.015 (0.011)	-0.007 (0.023)	0.045** (0.023)	0.053** (0.021)
Observations	1938	2144	2079	2050	2050
R^2	0.170	0.158	0.134	0.154	0.203
Adjusted R^2	0.116	0.110	0.083	0.104	0.156

Notes: Table A18 presents the association of the PGI score for cognitive skills on revealed preferences. The main specification controls in Panel A include gender, year of birth, age, age squared, the first ten principal components of the genetic data, parental educational attainment, occupational status, and an indicator for parental divorce. Panel B adds own education to the control set. Panel C adds log household net income to the main specification. Panel D combines all these controls. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Authors' calculations based on SOEP-IS 2020.

B.9. Weighting for Selective Consent and Adjusting for Sample Selectivity

In many empirical studies – especially those involving sensitive information such as genetic data – only a subset of individuals consent to participate. If the decision to consent is correlated with characteristics related to the outcome, the analysis sample may not be representative of the target population. This *selective consent* can introduce bias into the estimates if not properly addressed.

Estimation of Adjustment Weights. To correct for selective consent, we estimate adjustment weights that account for differences between those who consent and those who do not. The procedure is as follows:

- (a) **Selection Model:** We estimate a selection model (using a logistic regression) where the dependent variable is an indicator S_i that equals 1 if an individual is included in the analysis sample and 0 otherwise.
- (b) **Predictors:** The model includes a broad set of covariates, W_i , that are believed to influence the likelihood of consent. These covariates may include demographic characteristics, socioeconomic indicators, and other relevant attributes.
- (c) **Adjustment Factor:** For each observation, we predict the probability $\hat{p}_i = P(S_i = 1 | W_i)$ of being included in the analysis sample. The adjustment factor is then given by the inverse of this probability, $\frac{1}{\hat{p}_i}$.

This procedure extrapolates the analysis sample to represent the population of interest more accurately.

Caveats on Covariate Selection. A key consideration in this procedure is the choice of covariates W_i used in the selection model. In particular, one must be cautious about including covariates that are themselves outcomes (or post-treatment variables) of the explanatory variable in the primary analysis. If a covariate is affected by the treatment or explanatory variable, controlling for it in the selection model can lead to *post-treatment bias*. Such bias occurs because these variables lie on the causal pathway between the treatment and the outcome, and adjusting for them may remove part of the true effect or even induce spurious associations (Imbens and Rubin 2015; Angrist and Pischke 2008; Morgan and Winship 2014).

Thus, it is crucial that the covariates included in the selection model satisfy the following:

- They should predict participation (or consent) independently of the causal pathway between the primary explanatory variable and the outcome.
- They should be exogenous to the treatment effect – *i.e.*, not be influenced by the explanatory variable under study.

Failure to adhere to these principles may result in adjustment factors that inadvertently control for part of the effect one aims to measure, thereby distorting the estimated impact.

TABLE A19. Weighted association between PGI CS and risk/time preferences

	(1) Risk aversion	(2) Smoking	(3) Patience	(4) Fin. assets
PGI CS	0.048** (0.021)	-0.035*** (0.010)	0.004 (0.023)	0.032*** (0.011)
Observations	2298	2057	2200	2256
R ²	0.139	0.142	0.063	0.115

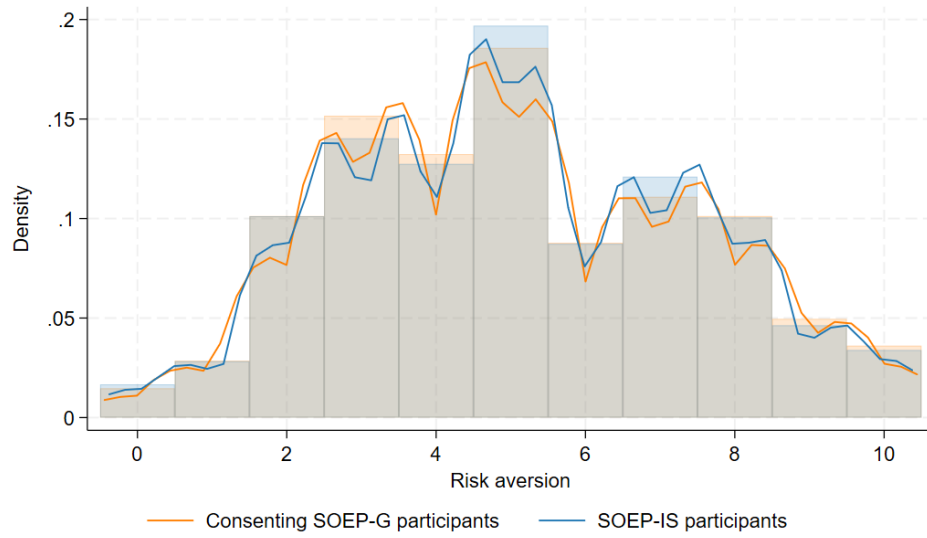
Notes: Table A19 displays the association between PGI score for cognitive skills and risk/time preference measures based on weighted OLS regressions. The weights are constructed based on the consent of SOEP-IS respondents to participate in the SOEP-G sample. The regression controls for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender and birth year. Robust standard errors are included in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

TABLE A20. Weighted association between PCI CS and social preferences

	(1) Trust	(2) Pos. reciprocity	(3) Neg. reciprocity	(4) Altruism	(5) Lending help	(6) Lending belongings	(7) Lending money
PGI CS	0.094*** (0.022)	-0.025 (0.024)	-0.058*** (0.022)	0.017 (0.023)	-0.008 (0.022)	0.053** (0.022)	0.055*** (0.021)
Observations	2189	2197	2197	2130	2173	2166	2167
R ²	0.122	0.059	0.080	0.106	0.132	0.139	0.209

Notes: Table A20 displays the association between PGI score for cognitive skills and social preference outcomes based on weighted OLS regressions. The weights are constructed based on the consent of SOEP-IS respondents to participate in the SOEP-G sample. The regression controls for age, age-squared, parental education and occupation, the first 10 principal components of genetic data, and indicators for growing up in a two-parent household, gender and birth year. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. *Source:* Authors' calculations based on SOEP-IS 2020.

FIGURE A3. Distributions of risk aversion



Notes: Figure A3 compares the distribution of risk aversion among SOEP-IS participants against that of the subset who consented to participate in SOEP-G, from which our analytical sample is drawn. The bars indicate the histogram of each sample overlying on top of one another. The lines are obtained from kernel density estimations using Epanechnikov kernel. For the SOEP-G sample coloured in orange, the bandwidth used is 0.450. The bandwidth in the estimation for the SOEP-IS, coloured in blue, is 0.398.

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